Universiteit Antwerpen
Faculteit Toegepaste Ingenieurswetenschappen

5 – Infrastructuur & Beveiliging

Lab 1: Introduction to OpenWrt and switching

Wout Van Uytsel
Femke Wouters

Faculteit Toegepaste Ingenieurswetenschappen
Campus Groenenborger
Gebouw g.US.103
Groenenborgerlaan 171
2020, Antwerpen

Wout.vanuytsel@uantwerpen.be
Femke.wouters@uantwerpen.be

Table of Contents

1 INTRODUCTION TO OPENWRT 3

2 DEVICES 4

3 LAB GOALS 5

4 EXERCISE 1: GETTING STARTED WITH THE GUI 5

5 EXERCISE 2: GETTING STARTED WITH THE COMMAND LINE 7

6 EXERCISE 3: PACKAGE MANAGEMENT 9

7 EXERCISE 4: CONNECTING TO YOUR NEIGHBOR 12

8 EXERCISE 5: STORING AND LOADING CONFIGURATIONS 13

REFERENCES ..... FOUT! BLADWIJZER NIET GEDEFINIEERD.

1 Introduction to OpenWrt

OpenWrt is a community-driven GNU/Linux operating system that targets embedded networking devices, most notably Wi-Fi routers and small edge gateways. The core idea is that, instead of treating a router as a fixed "firmware appliance," OpenWrt treats it as a small but fully manageable Linux system: you get a writable filesystem and a software ecosystem where features can be installed or removed through package management, rather than being permanently baked into one monolithic vendor image [1].

> **"This frees you from the restrictions of the application selection and configuration provided by the vendor and allows you to use packages to customize an embedded device to suit any application."**

Configuration in OpenWrt is intentionally structured for clarity and repeatability. Most services are configured through UCI (the Unified Configuration Interface), which centralizes settings and stores them in a consistent way (commonly under `/etc/config/`), making it easy to inspect, copy, and compare configurations across lab groups. For administration, the same configuration can be managed via SSH/CLI or via LuCI, the OpenWrt web interface, which is useful in a course setting because it visualizes networking constructs like interfaces, DHCP, and firewall zones while still mapping back to the same underlying config model.

1.1 Getting started with OpenWrt Devices

Routers and access points often ship with "fixed" vendor firmware: you get a fixed feature set and limited control. **OpenWrt** is different: it is a Linux-based OS for embedded networking devices that is explicitly designed to be **fully configurable and extensible** instead of being a locked firmware image.

Figure 1: OpenWrt logo

One of the nicest things about OpenWrt is that it turns a cheap router into a flexible, easy-to-manage networking platform. By simply flashing the firmware, you get a system that you can configure like a "real" network device and then extend with additional functionality when you need it. In practice, that means a single OpenWrt box can act as a switch (bridging and VLANs), a router (LAN/WAN interfaces, routing, DHCP/DNS), and a firewall (zone-based policy and filtering) without requiring specialized hardware or vendor licenses. That combination makes it extremely convenient for teaching, because students can explore how these components interact in one coherent environment instead of treating them as separate black boxes.

It's true that OpenWrt will generally not match the raw performance of dedicated enterprise gear like a Cisco switch or router. **Enterprise devices often have purpose-built switching ASICs and more powerful hardware optimized for high-throughput forwarding and large-scale feature sets, while OpenWrt commonly runs on consumer-grade CPUs and limited memory**. However, for learning, that trade-off is usually acceptable and sometimes even beneficial. **OpenWrt makes network behavior visible**: students can see configuration changes reflected immediately, verify the effect with practical tests, and troubleshoot using both a web UI and a command line. Because the configuration is explicit and repeatable, it also supports good engineering habits in a lab setting: documenting changes,

backing up configurations, restoring devices, and incrementally adding features (for example, VPNs or routing daemons) as the course progresses. And **this without requiring you to study vendor-specific commands.**

2 Devices

In this lab, we will be mainly using two types of devices: the **Netgear GS108T-300PES** and the **RT-AX53U**. We have already flashed these devices with the OpenWrt firmware, so every time the devices reset, you will return to a clean OpenWrt install (not the vendor firmware). If you would like to flash your own device, you can find more information about the procedure on the OpenWrt website [1].

> **NEVER remove the power from the device when it is rebooting or resetting, or you could permanently brick the device!**

2.1 Netgear GS108Tv3

In this lab, we use the **Netgear GS108T-300PES** (marketed as **GS108Tv3**) as a compact, managed Layer-2 switch. It is an **8-port 10/100/1000BASE-T (Gigabit Ethernet) Smart Managed Pro** switch in a small desktop form factor, designed for small networks where you still want control over segmentation and traffic behavior. The switch is fanless (silent) and intended for desktop or wall mounting [2]. In this course, we refer to the Netgear device as the switch since that is what it was originally designed to do.



Figure 2: Netgear 8 Port Switch

2.2 Asus RT-AX53U

Hardware-wise, the RT-AX53U is an **AX1800 dual-band Wi-Fi 6** device with **2×2 MIMO on 2.4 GHz and 5 GHz**, advertised PHY rates of **574 Mbps (2.4 GHz)** and **1201 Mbps (5 GHz)**, and support for Wi-Fi 6 mechanisms such as OFDMA and 1024-QAM. It has **four external antennas**, a **dual-core CPU**, **256 MB RAM**, and **128 MB flash**, which is plenty for the kinds of routing, VLAN, DHCP/DNS, and firewall labs we run (but it is not meant to compete with enterprise routers/switches in pure throughput or ASIC-accelerated switching). On the wired side, it provides **one Gigabit WAN port and three Gigabit LAN ports**, plus a **USB port** [3]. In this course, we refer to the Asus device as the router since that is what it was originally designed to do.



Figure 3: Asus RT-AX53U

3 Lab goals

Since this lab can feature students with different backgrounds, we will start the lab with a revision of some basic concepts. The goals of the lab today are the following:

- Access and navigate the OpenWrt LuCI web interface.
- Connect to an OpenWrt device using SSH and perform basic system inspection.
- Configure basic network connectivity.
- Compare the default networking behavior of an OpenWrt switch and router.
- Update OpenWrt package repositories and install software packages.
- Back up and restore OpenWrt configurations.
- Reset an OpenWrt device to its default state.

4 Exercise 1: Getting started with the GUI

4.1 Connect to the device's Graphical User Interface (GUI)

To explore OpenWrt in this lab, pair up with another student. One of you will connect your laptop to a switch, while the other connects to the router. For both devices, use an Ethernet connection to a LAN port (not the WAN/Internet port) and open the LuCI web interface in your browser. In a default OpenWrt configuration, the switch's and router's management interface is available at `http://192.168.1.1`. After connecting, compare your experiences and discuss how the IP address allocation differs between the switch and the router.

4.1.1 Student 1 – Connect to the switch

To access the switch's web interface:

1. Plug your laptop into one of the switch's ports.
2. Assign a **static IP** to your laptop within the same subnet as the switch's default management IP (for example, `192.168.1.10` with subnet mask `255.255.255.0`). Make sure this IP is unique on the network. Leave the gateway empty for now.

3. Open a browser and navigate to the switch's management IP at `http://192.168.1.1`
4. Log in to the LuCI interface using the default username root with **no password**. In this lab environment, you do not need to change these credentials, but never use defaults like this in a real deployment.

If you did these steps correctly, you should be greeted by a system status page.

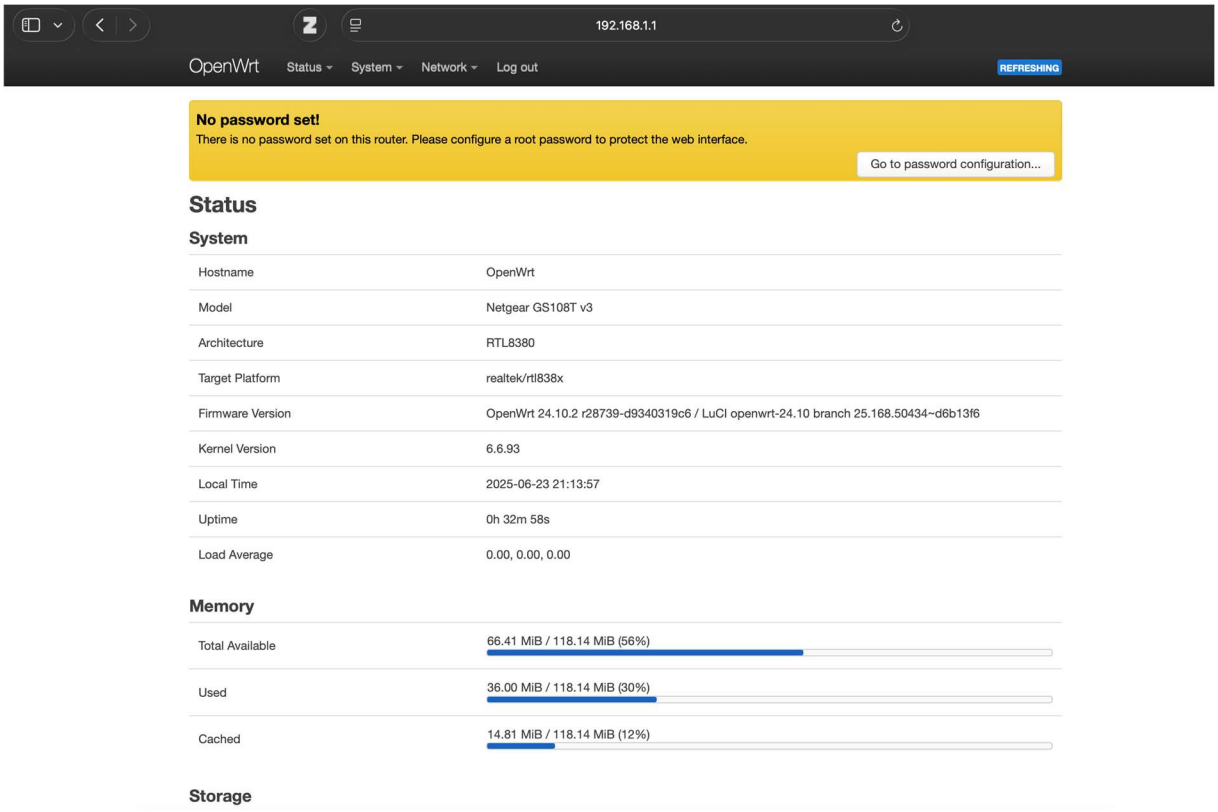


Figure 4: OpenWrt starting screen for the switch

4.1.2 Student 2 – Connect to the router

By default, OpenWrt runs a DHCP server on the router's LAN interface, so your laptop will automatically receive an IP address in the same subnet. While the switches are also capable of running a DHCP server, we do not enable it in this lab because it consumes too many resources. The router, however, does provide DHCP, allowing us to test automatic IP assignment. To access the router's web interface:

1. Plug your laptop into one of the router's LAN ports.
2. Ensure your laptop is set to **obtain an IP automatically via DHCP**. Within a few seconds, it should receive an IP in the router's subnet (`192.168.1.x`), and you will see that the default gateway is set to the router's LAN IP (`192.168.1.1`).
3. Open a browser and navigate to the router's management IP at `http://192.168.1.1`.
4. Log in to the LuCI interface using the default username root with no password. In this lab environment, you do not need to change these credentials, but never use defaults like this in a real deployment.

> Tip: If your laptop does not receive an IP, you can temporarily assign a static IP within the router's subnet to access the GUI, then return to DHCP.

If you did these steps correctly, you should be greeted by a system status page.

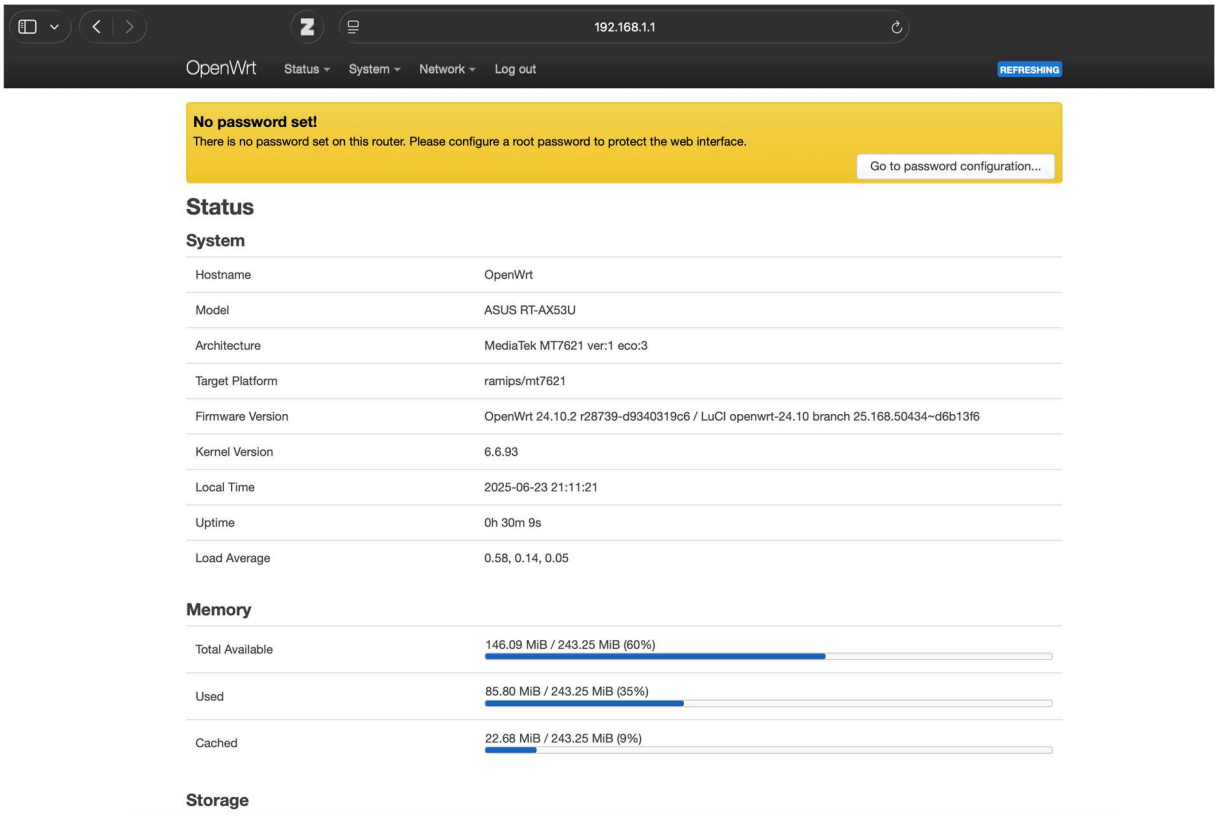


Figure 5: OpenWrt starting screen for the router

4.2 Explore the GUI

Explore the different tabs and menus, what you can find where, and what is currently there in the standard config.

5 Exercise 2: Getting started with the command line

5.1 Connect to the device through the command line

While connecting to the GUI is one step, you will need command access to the device as well. Not everything is accessible through the GUI, and therefore, command-line access is needed.

Since the next steps are the same for both the switch and the router, you can choose which device to work with. Connect your laptop to a switch port or to a LAN port on the router, and access the device via SSH. On a default OpenWrt setup, the router is typically reachable at `192.168.1.1`. The SSH username is normally **`root`**. The command to connect on Windows and Mac is the following:

```
ssh root@192.168.1.1
```

On the **first** connection, your SSH client will ask you to confirm the device's host key (a normal security step). Type **`yes`**.

To end the SSH session:

```
exit
```

If you get a “host identification has changed” warning (common after reflashing OpenWrt or swapping devices), remove the old key entry and try again:

```
ssh-keygen -R 192.168.1.1
ssh root@192.168.1.1
```

If you are correctly connected, you should see the following appear:



Figure 6: OpenWrt SSH

5.2 OpenWrt CLI warm-up

When you SSH into OpenWrt, you land in a small Linux shell. Unlike Cisco IOS, there is no “user vs privileged” mode; you are typically logged in as `root`. This root access means that you can change everything on the device, so be thoughtful about the changes you make.

In this exercise, we will try some commands. Find out what the output is and what the output means.

Command	Output and function
`ubus call system board`	…
`cat /etc/openwrt_release`	…
`uptime`	…
`free -h`	…
`df -h`	…
`ip a`	…
`ip route`	…
`ifstatus lan`	…

`ifstatus wan`	…
`uci show network`	…
`logread`	…
`logread -f`	…
`logread -e dnsmasq`	…
`logread -e firewall`	…

Connectivity commands that can be useful throughout the lab.

Command	Output and function
`ping -c 4 XXX.XXX.XXX.XXX`	…
`ping 8.8.8.8`	…
`nslookup openwrt.org`	…
`fping -g 10.1.1.0/24`	…

6 Exercise 3: Package management

In this exercise, you will learn how to **update your switch or router** and **install necessary software packages**. Keeping devices updated ensures they have the latest features, bug fixes, and security patches. OpenWrt (version 24.10 and older) uses the **`opkg` package manager**, which handles downloading, installing, and updating software packages.

Before starting, connect to your device **via the command line using SSH** from your laptop, as described in the previous exercise.

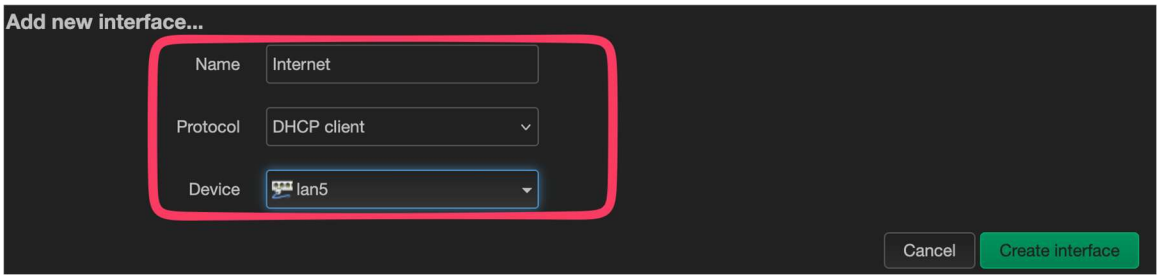
6.1 Connect to the internet.

The first important step is to **connect your switch or router to the Internet**. This connection is required for the device to download package lists and install new software. For the Asus router, plug the cable to the campus network into the WAN port.

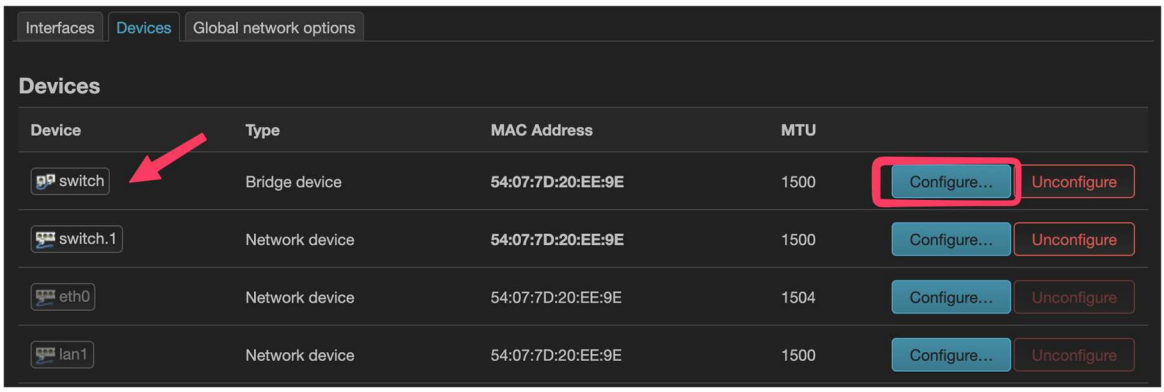
To connect the Netgear switch to the Internet, it requires a few extra steps.

1. Plug in the cable to the campus network into one of the free ports of the device.

2. In the GUI, go to **Network → Interfaces** and, under the Interfaces tab, press "Add new interface…". Choose the port you have plugged the cable into and choose DHCP Client as the protocol.



3. Afterwards, go to the Devices tab, click Configure…, and remove the port from the bridge device.



4. Save and apply the settings.

Test your internet connection by sending a ping to a website. You can also do this in the LuCI web interface by navigating to **Network → Diagnostics** and entering openwrt.org in the ping field. If you receive a response, your router is successfully connected to the Internet.

> **Important: Once you have finished the installation and confirmed that everything is working, always unplug the device from the Internet to prevent unintended Internet access.**

6.1.1 Check the system date.

OpenWrt uses **HTTPS to download package lists**. If the system date is incorrect, SSL certificates may appear invalid, causing errors like:

```
wget: SSL verify error: unknown error
opkg_download: wget returned 5
```

SSL certificates used by the OpenWrt package repository have **validity periods**. If your device clock is wrong, HTTPS verification will fail, preventing `opkg update` from downloading package lists. Correcting the date ensures that your device can securely connect to the repository and install software. Check the current date:

```
date
```

If incorrect, you can fix it manually by executing the following command. Use today's date:

```
date -s "2026-03-10 14:00:00"
```

6.2 Updating the package lists

Before installing or upgrading any software, you need to update the local package lists. This ensures your device knows about the latest available versions:

```
opkg update
```

This command fetches the latest package information from the repositories configured on your device. Always run this command before installing new software to avoid errors or outdated installations.

6.3 Upgrading installed packages

If you need to upgrade existing packages, you can check which packages have updates available:

```
opkg list-upgradable
```

If the returned list is empty, it simply means that all installed packages on your OpenWrt device are already up to date. If the list is not empty, you can upgrade the packages using:

```
opkg upgrade <packagename>
```

6.4 Installing new packages

To install a new package, run:

```
opkg update && opkg install <packagename>
```

For example, to install networking tools:

```
opkg update && opkg install tcpdump
```

This command first updates the package lists, then downloads and installs the specified packages.

After installation, you can check that the package is installed using:

```
opkg list-installed | grep <packagename>
```

For example:

```
opkg list-installed | grep tcpdump
```

6.5 Installing packages via the GUI

OpenWrt also allows package management through its web interface, **LuCI**, which can be easier for users who prefer a graphical approach. After logging in, navigate to **System → Software**. Here, you can click **Update lists** to refresh the available packages. You can explore the **Available packages** to see what can be installed, and the **Installed packages** section to see what is already present on the device. Use the filter box in the Installed packages section to confirm that `tcpdump` was successfully installed.

To try installing another package via LuCI, search for a known package such as `vim`. Once you find it, click **Install**. LuCI will download and install the package, and it will then appear in the Installed packages list. You can also remove or reinstall any package with a single click, making LuCI a convenient alternative to the command line for package management.

7 Exercise 4: Connecting to your neighbor

In this exercise, we are going to explore the layer two connectivity. Currently, the network is very isolated, and therefore, the IP choice is not too important. However, the setups will get increasingly complicated, which requires us to do good housekeeping.

In the following setup, we want to connect two hosts by using one switch and one router, both acting as Layer 2 devices. All these devices are in the same network (`/16`). You will configure **static IP addressing everywhere**. Do not connect the two devices to each other until both IPs are changed

For the first labs that are not focused on security, we will turn off the firewall on all OpenWrt devices with the following command.

```
/etc/init.d/firewall stop
```

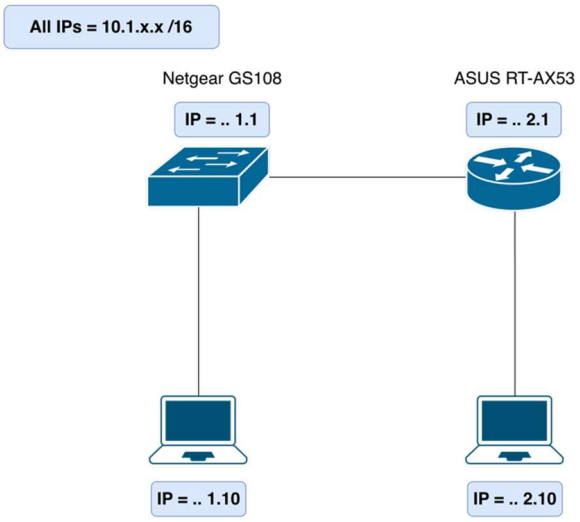


Figure 7: Exercise 4 setup

Start by configuring the devices. Connect your laptop to a LAN port of the device and open the LuCI web interface. On a default OpenWrt installation, the LAN management IP is `192.168.1.1`, so you should be able to browse to that address initially. Once logged in, navigate to **Network → Interfaces → LAN → Edit** and change the **IPv4 address** to **(see diagram)** and the **netmask** to `255.255.0.0 (/16)`. Now click *Save & Apply.*

What will happen to your browser connection the moment you change the router's IP address, and what exact URL will you need to use afterwards to reach LuCI again?

_____

_____

What changed on your laptop side when the IP changed? Did your laptop remain on the correct network to reach `10.1.x.1`?

_____

_____

Now configure the two hosts with static IPs. Give the host on the switch side `10.1.1.10/16`, and give the host on the router side `10.1.2.10/16`. And physically cable the setup exactly like the figure:

a host connected to the switch, a host connected to the OpenWrt router, and one Ethernet link between the switch and one of the router's LAN ports.

Since both hosts are in `10.1.0.0/16`, do you need a default gateway for host-to-host connectivity inside this lab network?

_____

_____

At this point, ask yourself: if everything must live inside one `/16` subnet, do you want the OpenWrt WAN port involved at all, or should this be LAN-to-LAN? Why?

_____

_____

Finally, verify connectivity and management access. From each host, test reachability to both management IPs and to the other host. You should be able to open the management GUIs in a browser at `http://10.1.1.1` (Netgear) and `http://10.1.2.1` (Asus). If you cannot, stop and diagnose systematically: *does the host have the correct IP and* `/16` *mask, is the cable in the expected port, and are you trying to access the old management IP after you changed it?*

Did you encounter any problems along the way, and how did you solve them?

_____

_____

8 Exercise 5: Storing and loading configurations

In this exercise, you learn three operational basics for working with OpenWrt in a lab: (1) exporting a backup of the current configuration, (2) restoring a previously saved configuration archive, and (3) resetting the device back to a clean default state using the physical reset button. The goal is that you can always recover from mistakes and return to a known-good configuration without starting from scratch.

8.1 Back up (download) your current configuration

Connect to the OpenWrt web interface (LuCI) of your device.

1. Navigate to **System → Backup / Flash Firmware**.
2. Click **Generate archive** to download a configuration backup (typically a `*.tar.gz`).
3. Save the file on your computer and **rename it clearly**, for example: `Group<n>_DeviceA_OpenWrt_24.10.2_backup_YYYY-MM-DD.tar.gz`

What you just downloaded is a configuration archive (not a full firmware image). In practice, it captures your device configuration files (commonly the contents under `/etc/`, including `/etc/config/`), and LuCI also allows including additional paths via the configuration list on that page.

Try downloading your current configuration.

8.2 Restore (upload) a previous configuration archive

To upload your configuration, do the following:

1. Navigate again to **System → Backup / Flash Firmware**.
2. In the **Restore backup** section, select your previously saved `*.tar.gz` configuration archive and upload it.
3. Confirm the restore. The device will apply the configuration and may reboot.

Important operational consequence: if the restored configuration contains different **IP addressing** (e.g., a different management IP), you will "lose" the web UI temporarily simply because your browser is still trying the old address. In that case, reconnect using the **management IP that is defined inside the restored configuration**.

Upload the configuration file you just downloaded.

8.3 Resetting the device

When the lab ends or if you have problems with your device that you cannot fix, you can always reset the device back to its default clean image. This will remove the packages, and it will go back to its original settings. You will have to do this at the end of every lab or in case of problems.

To reset the device back to a clean default state using the hardware button:

1. Ensure the device is powered on.
2. **Press and hold the reset button for about 10 seconds**, then release it.
3. The device should perform a factory reset and reboot. **DO NOT UNPLUG THE DEVICE NOW!**

After a factory reset, you should expect your custom settings to be cleared. On OpenWrt, this is typically implemented by clearing the writable overlay (the part where configuration changes and installed packages live), which effectively returns the system to its initial state.

If a device's reset button does not behave as expected or you prefer to do it in the software, LuCI also provides a "Reset to defaults / Perform reset" option on the same System → Backup / Flash Firmware page, which triggers the same "return to defaults" outcome via software.

At the end of the lab, reset the device, wait for it to finish rebooting, and put everything back in the correct boxes.

What is the difference between a configuration backup archive and a firmware image?

_____

_____

After restoring the backup, why might you temporarily lose access to the device even though the restore action succeeded? Do you need to change anything on your hosts?

_____

_____